Improving Supply Chain Transparency Through Blockchain: The Volvo Cars – Circulor Partnership

Digital Assignment 1

Submitted by

Divya Ranjith (23BLC1291)

Shambhavi (23BLC1226)

Smruthi (23BLC1249)

A Case study report submitted to

Dr. Dhivya R

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING (SCOPE)

in partial fulfilment of the requirements for the course of

BCSE324L Foundations of Blockchain Technology

in

B.Tech. ELECTRONICS AND COMPUTER ENGINEERING

July 2026

TABLE OF CONTENTS

S. No.	Title	Page No.
1	Introduction	3
2	Background of the Organizations / Industries	3
3	Problem Addressed	4
4	Blockchain Solution Implemented	5
5	Benefits and Outcomes	6
6	Challenges and Limitations	6
7	Future Scope	7
8	Conclusion	8
9	References	8

I. INTRODUCTION

Blockchain technology improves security, transparency, and trust by using a decentralized ledger shared among authorized participants. Unlike traditional databases, blockchain stores tamper-resistant records that are securely linked through cryptography, making data reliable and easy to verify. Its applications now extend beyond cryptocurrencies to industries such as finance, healthcare, and supply chain management.

In supply chains, blockchain enhances traceability by providing a secure, shared record of transactions among suppliers, manufacturers, distributors, and regulators. This reduces issues such as counterfeit products, fraud, and poor visibility while improving accountability and regulatory compliance.

The rapid growth of electric vehicles (EVs) has increased the need for transparent battery supply chains. Regulations such as the European Union Battery Regulation require digital Battery Passports for EV batteries sold in the EU from 2027, containing information on material origin, recycled content, carbon footprint, and lifecycle history.

To meet these requirements, Volvo Cars partnered with Circulor to develop a blockchain-based Battery Passport for the Volvo EX90. The system tracks raw material origin, supply chain movement, recycled content, manufacturing details, and carbon emissions, providing greater transparency and supporting sustainability compliance.

This report presents a case study of the Volvo Cars–Circulor battery traceability solution, examining its implementation, benefits, limitations, and role in creating transparent and sustainable EV supply chains.

II. BACKGROUND OF THE ORGANIZATIONS / INDUSTRIES

Volvo Cars

Volvo Cars is a Swedish automobile manufacturer founded in 1927 and headquartered in Gothenburg, Sweden. The company is recognized for its focus on safety, innovation, and sustainability. It aims to become a fully electric vehicle manufacturer by 2030 and achieve net-zero greenhouse gas emissions across its value chain by 2040. Volvo adopted blockchain technology to improve the traceability of lithium-ion battery materials. This enables responsible sourcing, greater supply chain transparency, and compliance with evolving sustainability regulations.

Circulor

Circulor is a UK-based technology company founded in 2017 that provides supply chain transparency solutions using permissioned blockchain. Its platform tracks raw materials from mining to the finished product, recording information such as material origin, ownership, carbon emissions, and compliance with responsible sourcing standards.

In 2020, Volvo Cars invested in Circulor through the Volvo Cars Tech Fund, strengthening their partnership in developing transparent and sustainable battery supply chains.

Electric Vehicle Battery Industry

The growing demand for electric vehicles has increased the need for transparent lithium-ion battery supply chains. Critical minerals such as lithium, cobalt, nickel, and graphite pass through multiple stages and organizations, making traceability difficult. Disconnected systems and limited visibility can reduce data reliability and make responsible sourcing harder to verify.

To address these challenges, the European Union Battery Regulation requires digital Battery Passports for EV batteries sold in the EU from 2027. In response, Volvo Cars and Circulor developed a blockchain-enabled Battery Passport that securely tracks battery information, improves transparency, and supports regulatory compliance.

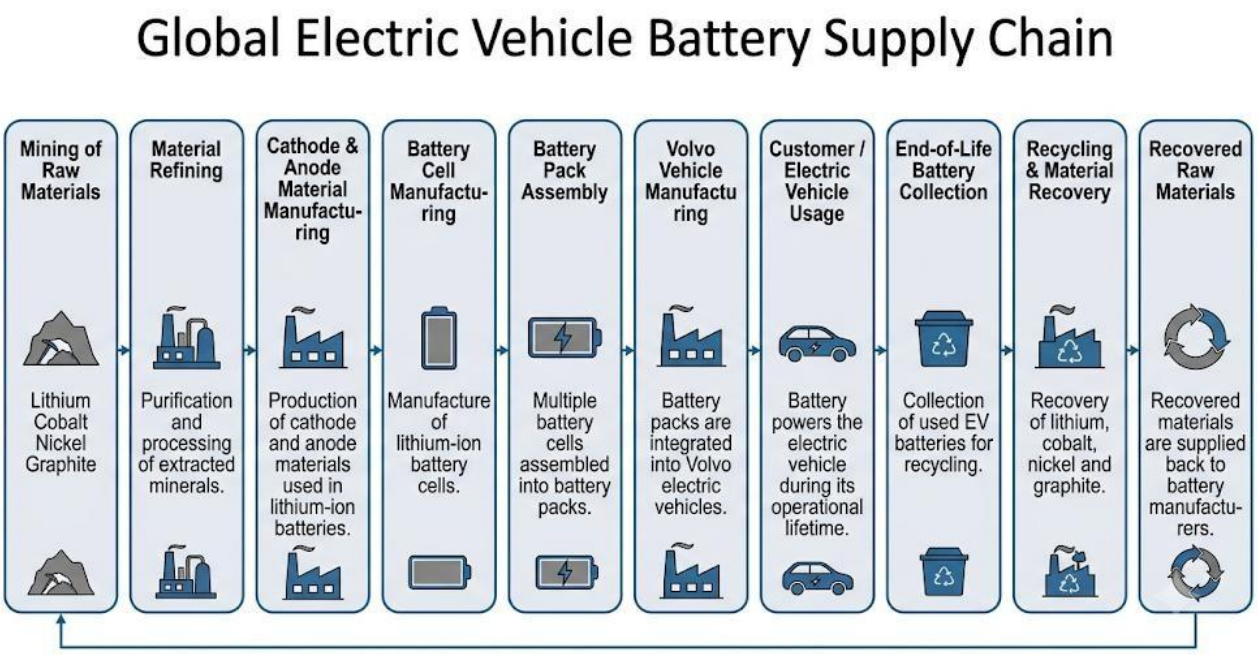


Figure 1. Global electric vehicle battery supply chain showing the lifecycle of critical battery materials from mining to recycling and reuse.

III. PROBLEM ADDRESSED

The rapid growth of the electric vehicle (EV) industry has made battery supply chains more complex, involving multiple organizations and countries. Critical materials such as lithium, cobalt, nickel, and graphite are difficult to track because supply chain information is stored in separate systems, limiting transparency and traceability.

Ethical sourcing, especially of cobalt from the Democratic Republic of the Congo (DRC), is another major challenge. Traditional systems rely on fragmented databases and manual records, making it difficult to verify supplier information and prevent fraud.

At the same time, regulations such as the European Union Battery Regulation require Digital Battery Passports for EV batteries sold in the EU from 2027. Manufacturers must provide accurate information on material origin, recycled content, carbon emissions, and battery lifecycle, which conventional systems struggle to support.

To overcome these challenges, Volvo Cars partnered with Circulor to implement a blockchain-based traceability platform. The solution improves transparency, ensures secure and tamper-resistant records, supports responsible sourcing, and helps meet regulatory requirements.

Traditional Supply Chain	Blockchain-Based Supply Chain
Centralized records	Distributed ledger
Manual verification	Shared verification
Limited traceability	End-to-end traceability
Easy data modification	Immutable records

IV. BLOCKCHAIN SOLUTION IMPLEMENTED

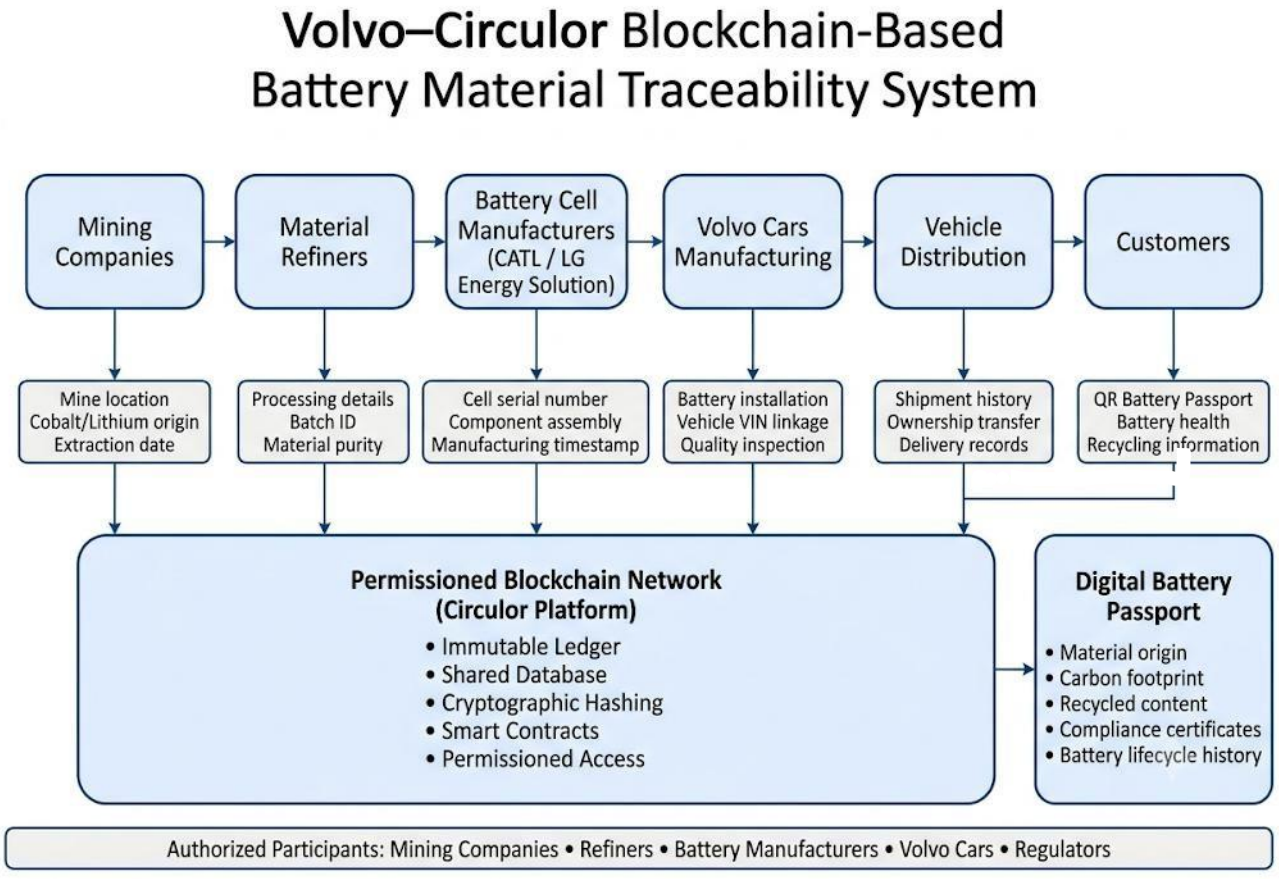
To improve battery traceability and meet sustainability requirements, Volvo Cars partnered with Circulor to implement a permissioned blockchain platform across its EV battery supply chain. Initially developed to trace cobalt, the system was later expanded to track other critical materials such as lithium and mica. Unlike public blockchains, only authorized participants can access and validate data, ensuring secure and controlled information sharing.

The platform connects mining companies, suppliers, battery manufacturers, logistics providers, Volvo Cars, and regulators. At each stage, verified information, including material origin, ownership, batch details, transportation records, carbon emissions, and compliance certificates, is stored on a shared, tamper-resistant ledger. This creates a reliable chain of custody throughout the battery lifecycle.

Instead of replacing existing systems, Circulor integrates with them through APIs, allowing organizations to securely upload verified data while minimizing changes to their operations. Role-based access control ensures each participant can access only the information required for their role. For example, suppliers update production records, Volvo verifies material traceability, regulators review compliance information, and customers access selected Battery Passport details through a QR code.

To improve data reliability, the platform combines blockchain with AI-based analytics and supplier audits to identify inconsistencies and verify submitted information. This addresses the limitation that blockchain protects stored records but cannot confirm the accuracy of data at the time it is entered.

The platform has evolved into the Digital Battery Passport for the Volvo EX90. It provides verified information on material origin, recycled content, carbon footprint, manufacturing history, and battery lifecycle. The solution improves transparency, supports regulatory compliance, and promotes sustainable battery management.



V. BENEFITS AND OUTCOMES

Circulor's blockchain platform has greatly improved the traceability of Volvo Cars' battery supply chain. It allows critical materials such as cobalt and lithium to be tracked from mining to battery production, vehicle assembly, and recycling. This end to end visibility helps Volvo verify material origin, prevent unethical sourcing, and improve supply chain transparency.

The platform also supports responsible sourcing by providing secure, tamper resistant records of supplier information, compliance certificates, and material history. This increases trust in the accuracy of supply chain data and helps Volvo meet its environmental, social, and governance (ESG) goals.

Another key benefit is easier regulatory compliance. The system stores verified information on material origin, recycled content, carbon footprint, and manufacturing history, helping Volvo meet the European Union Battery Regulation and support Digital Battery Passports with less manual effort.

By replacing fragmented records with a shared source of information, the platform improves collaboration between supply chain partners and speeds up data verification. The Digital Battery Passport in the Volvo EX90 also allows customers to access selected battery information through a QR code, while regulators and recyclers can use reliable lifecycle data. Overall, the solution strengthens transparency, builds stakeholder trust, and supports sustainable battery management.

Benefit	Outcome for Volvo Cars
End-to-end traceability	Tracks battery materials throughout the supply chain
Ethical sourcing	Verifies responsible mineral procurement
Regulatory compliance	Supports EU Digital Battery Passport requirements
Transparency	Improves trust among stakeholders and customers

VI. CHALLENGES AND LIMITATIONS

Although blockchain offers many benefits, it also has several limitations. While it protects recorded data from being changed, it cannot verify whether the information entered is accurate. Incorrect data can still be permanently stored, making supplier verification and regular audits essential.

Integrating blockchain with existing systems is another challenge because organizations use different software, data formats, and processes. Smaller suppliers may also lack the technical resources needed to adopt the technology, making implementation more difficult.

Data privacy is also a concern. Even with permissioned blockchain networks, companies may hesitate to share sensitive business information. Strong access controls are needed to balance transparency with confidentiality.

In addition, implementing a blockchain network requires significant investment in infrastructure, system integration, training, and maintenance. As more participants and transactions are added, maintaining performance becomes increasingly challenging. Long term success therefore depends on industry collaboration, standardized data formats, and supportive regulations.

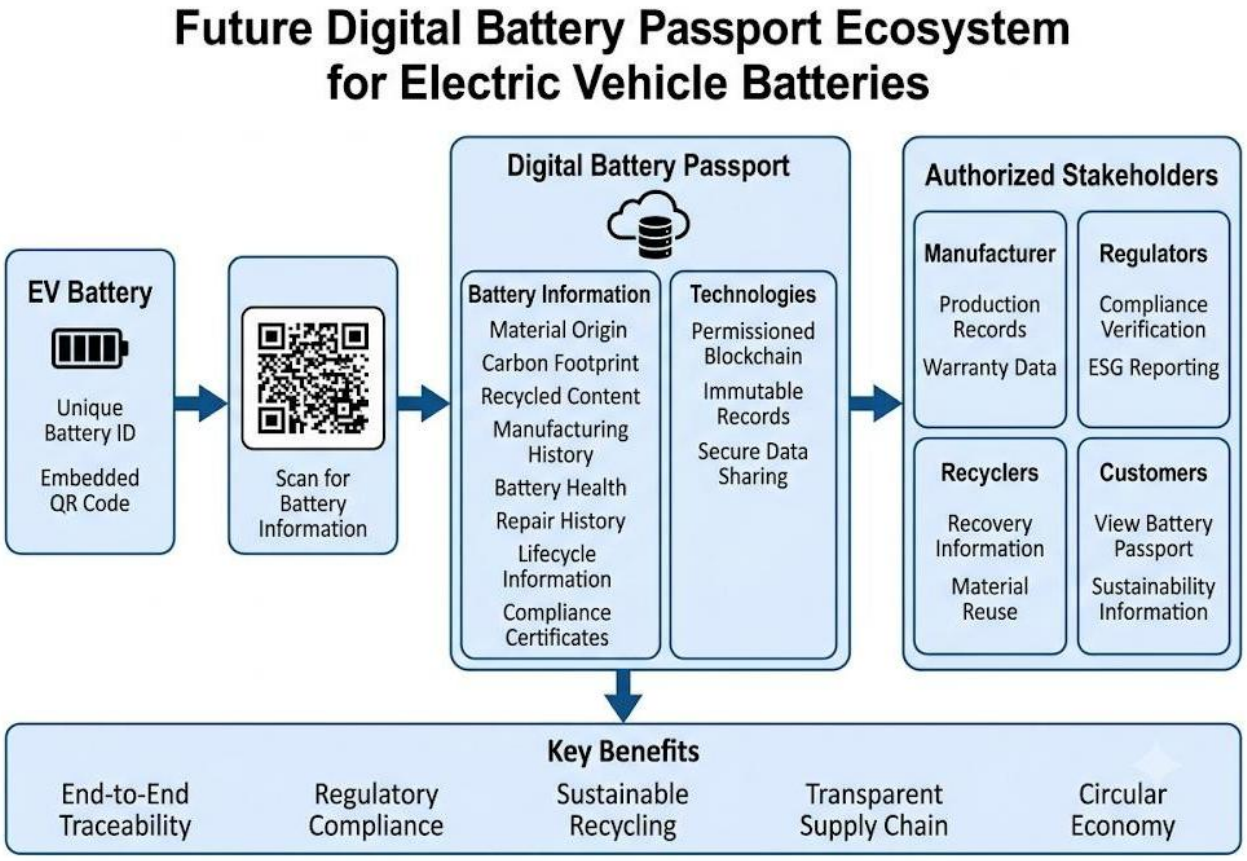
VII. FUTURE SCOPE

The future of blockchain in EV battery supply chains will focus on improving lifecycle management and supporting a circular economy. As EV adoption grows, blockchain systems can expand to track more materials, monitor environmental impacts, and improve collaboration between suppliers, manufacturers, and recycling organizations.

A major development will be the wider use of Digital Battery Passports. These records will store important information such as material origin, carbon footprint, recycled content, repair history, and end-of-life processes. Blockchain can provide a secure and reliable platform for sharing this data among manufacturers, regulators, recyclers, and customers.

Future systems are also expected to combine blockchain with technologies such as IoT, artificial intelligence (AI), and data analytics. IoT devices can collect real-time battery information, while AI can analyze data to improve maintenance, predict battery performance, and optimize supply chain decisions.

For widespread adoption, global collaboration and common data standards will be necessary. With greater industry participation, blockchain can support transparent sourcing, regulatory compliance, efficient recycling, and a more sustainable electric vehicle ecosystem.



VIII. CONCLUSION

The Volvo Cars and Circulor case study show how blockchain can improve transparency, traceability, and sustainability in electric vehicle battery supply chains. The permissioned blockchain platform enables secure tracking of critical materials from sourcing to manufacturing and recycling, helping Volvo verify responsible sourcing and reduce inaccurate supply chain information.

The solution also supports compliance with regulations such as the European Union Battery Regulation through Digital Battery Passports. By improving data sharing among manufacturers, suppliers, recyclers, and regulators, it contributes to more efficient and sustainable battery lifecycle management.

Although challenges such as data accuracy, system integration, cost, and adoption remain, the advantages of improved traceability and transparency make blockchain a valuable solution. As EV adoption increases, blockchain will continue to support ethical sourcing, circular economy practices, and digital transformation in the automotive industry. The Volvo and Circulor partnership demonstrates the practical potential of blockchain in solving real-world supply chain challenges.

IX. REFERENCES

[1] Volvo Cars, "Volvo Cars to implement blockchain traceability of cobalt used in electric car batteries," Nov. 6, 2019. [Online]. Available: https://www.volvocars.com/uk/media/press-releases/D1FC13B168C968C0/. [Accessed: Jul. 28, 2026].

[2] Circulor, "Digital Product Passports with Supply Chain Trust & Assurance," 2026. [Online]. Available: https://circulor.com/. [Accessed: Jul. 28, 2026].

[3] European Parliament and the Council of the European Union, *Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries*, Official Journal of the European Union, Jul. 2023.

[4] Reuters, "Volvo to issue world's first EV battery passport ahead of EU rules," Jun. 4, 2024.

[5] M. Casino, T. K. Dasaklis, and C. Patsakis, "A systematic literature review of blockchain-based applications: Current status, classification and open issues," *Telematics and Informatics*, vol. 36, pp. 55–81, 2019.

[6] F. Tian, "An agri-food supply chain traceability system for China based on RFID & blockchain technology," in *Proc. 13th Int. Conf. Service Systems and Service Management (ICSSSM)*, Kunming, China, 2016, pp. 1–6.

[7] M. Saberi, M. Kouhizadeh, J. Sarkis, and L. Shen, "Blockchain technology and its relationships to sustainable supply chain management," *International Journal of Production Research*, vol. 57, no. 7, pp. 2117–2135, 2019.

[8] K. Salah, M. H. U. Rehman, N. Nizamuddin, and A. Al-Fuqaha, "Blockchain for AI: Review and open research challenges," *IEEE Access*, vol. 7, pp. 10127–10149, 2019.

[9] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf

[10] Linux Foundation, "Hyperledger Fabric Documentation," 2024. [Online]. Available: https://hyperledger-fabric.readthedocs.io/